Four Questions, Four Answers: the Calibration Meets the Routing Theory

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This note answers the four open empirical questions: why myopia wins at the calibration and what flips it, whether the flat parameters are an artifact of our estimation, what Xinyuan's routing theory says when evaluated directly at the calibrated primitives, and the status of the two numerical properties the theory documents asked us to verify. The last of these produced counterexamples that change the theory agenda, so it comes with its scope carefully certified.

1 Boosting ρ: the flip exists but is capped

Meichun's question was whether raising ρ counterfactually restores the planning premium. It does, monotonically, and it also shows why ρ alone was never the full story: the aware-minus-myopic gap runs from 0.04% at the calibrated $\rho = 0.035$ (statistically zero) to 0.99% at $\rho = 1$. At $\rho = 1$ we also ran Xinyuan's primitive route as a third policy, recommending along the shortest path and harvesting: it captures 0.89% against myopia, nearly the full aware premium, which is direct evidence that the route is a near-optimal policy wherever cultivation has value at all. But one percent at $\rho = 1$ is the cap at our κ, which leads to the sharper diagnosis.

2 The phase map: κ is the gate, not the flat intercepts

Crossing intercept-spread multipliers $\{1, 3, 6\}$ with $\kappa \in \{1.966, 4, 6\}$ at $\rho = 0.45$ revises the diagnosis we converged on: stretching the intercept spread sixfold moves the gap from 0.44% only to 0.76%, while raising κ to 6 at the original spread moves it to 21.2%. The near-uniform consumption probabilities are real, but they are near-uniform because κ is small, not because the intercepts are close; making the intercepts wild changes almost nothing while making alignment discriminating changes everything. Every earlier result (the iso-engagement gate, the dense- geometry null, the backfire complementarity appearing only at $\kappa = 5$) lines up behind the same single parameter.

3 The estimation is not forcing the data

Regenerating click labels on the real design matrix from known truths, with intercept spreads up to six times ours and κ up to 6, all recentred to MIND's 4% click rate, the production estimation recovers κ to within 0.022 and the intercept profile with correlation at least 0.995 in every one of nine configurations. If tastes on this platform were heterogeneous and alignment-sensitive, our pipeline would have said so. The flat calibration is a property of news reading, not of the method. (Scope: this certifies the logit estimation stack conditional on the state construction; the state construction itself was validated separately by the placebo, ordering, and rolling-day designs.)

4 The routing theory evaluated at the calibration

The $\rho = 1$ theory is directly computable on our 14-category graph, and it predicts our null twice over. First, the skip inequality $1/p_{j\to k} + 1/p_{k\to i} - m_k/\bar{r}_{a^\star} \le 1/p_{j\to i}$ holds for *none* of the 2,184 ordered triples at the calibrated parameters, with a minimum slack of 17: no stepping stone pays, so the optimal route is direct harvesting, which is myopia. Under the dense content Gram the geometry does admit 51 profitable triples, but the second prediction closes that door too: the turnpike threshold is $\bar{H} \approx 167$ (one-hot) and ≈ 137 (dense) against 30-period sessions, so the theory itself says sessions this short play myopically. Patience breakpoints tell the same story in discounted form: the optimal map departs from myopia only above retention $\delta \approx 0.60$ in the $\rho = 1$ idealization, and far later once ρ is realistic. One honesty note on the frontier: with one-hot features the full dominance vector makes every item undominated for structural reasons, so the meaningful statement is the (α, m)-restricted frontier, which keeps {music, lifestyle, health, finance}; and the artificial $\alpha = -10$ test confirms the documented non-tightness, since an item nobody would ever click joins the frontier on margin alone.

5 The two conjectured properties fail, in scope

The documents flag two properties as numerically observed but unproven: the ordered-marginal-values inequality of Remark 7, and single crossing of action comparisons in the horizon. We ran 2,000 random instances ($n \in \{3, 5\}$, exact node recursion, $H = 60$), of which 1951 satisfy Assumption 2 in both its stated forms. Within scope, the upper half of the ordered-marginals inequality fails in 540 instances and the full two-sided version in 1606; single crossing fails at a comparable rate (566 instances overall), with up to 20 sign changes in one instance. The violations are not endgame artifacts (they occur deep in the turnpike regime, at horizons where the optimal map already equals the shortest-path map everywhere) and not numerics (magnitudes sit nine orders above float noise and survive long-double recomputation; an independent audit reproduced all counts exactly and verified the saved examples by separate implementation). Three complete counterexample instances, each certified to satisfy Assumption 2, are in the shared results file for Xinyuan. The constructive reading: the lemma is false in general, so the exact ladder and the interval conjecture need a primitive condition that defines a subclass, and finding that condition is now a concrete, example-guided task rather than an open-ended hope.

6 Items 4 and 5, revised

Resistance. The precise evidence status, reconciling the discussion: backfire exists and is small, which is exactly the configuration Xinyuan's theory needs. On the full impression scale, jointly estimated with ρ, the effect is decisively present (the simplex-native proportional-redistribution form is the data-preferred specification, likelihood ratio 1,489), with a per-impression magnitude of 0.002 to 0.01, attention-diluted, and far too small to threaten the routing structure. Its *policy* value remains counterfactual: material only when paired with $\kappa \ge 5$. Proposal: adopt resistance in the model as a perturbation term with the proportional form, treat the no-resistance case as the baseline the theory solves, and state the empirical magnitude as justification for the smallness assumption.

Index policy. The primitive route is the principled potential the index was missing. At $\rho = 1$ the route policy captures nearly the full premium; in the learning benchmark the geometric potential already reached 97% of oracle while the myopic-payoff potential failed, and the route value W now gives a potential with a certified error bound instead of a guess. The benchmark to beat remains estimate-then-plug-in planning, which is near-oracle whenever a few persistent intercepts are the only unknowns; the index's case should be built where plug-in struggles, with state uncertainty or short-lived items, and the route-based Φ is the right foundation for it.